Activity 09 - Analyze Vertex AI Hyperparameter Trials

Course: Google Professional Machine Learning Engineer Training Course code: C740 Version: v2.0 | 12 September 2026 Evidence: activity artifacts

Objective

Define a Vizier-style search space and select a trial using quality and cost evidence.

Deliverables

- trials.csv
- tuning_config.yaml
- select_trial.py
- A completed evidence/ folder with decision notes and verification output

Prerequisites

- Python 3.11 or later and a terminal; Docker only where the activity explicitly uses it.
- For optional live Google Cloud steps: the safe training project created in Activity 01.
- Start in this activity folder so relative paths and cleanup commands remain scoped.

Safety, cost and privacy

- Use a training project that contains no production or personal data.
- Replace every placeholder such as <PROJECT_ID> and <REGION> before a cloud command.
- Do not paste API keys or service-account JSON into notebooks, scripts or screenshots.
- Prefer Cloud Shell and Application Default Credentials. Never commit .env or credentials.
- Verify the result against the acceptance checklist, then run the cleanup section.

Evidence to retain

Create an evidence/ folder locally. Save command output, job or revision IDs, metric tables, screenshots without secrets, and a short decision note. Use filenames that include the activity number and date. Evidence is part of the learning outcome, not an optional extra.

Detailed step-by-step guide

Step 1 - Review tuning_config.yaml; identify search distributions, objective direction, maximum trials and parallel trials

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Check: Record the observable result or explain the error before continuing.

Step 2 - Inspect trials.csv for status, objective, duration and cost

Inspect trials.csv for status, objective, duration and cost. Exclude failed trials from the quality ranking but retain them for reliability analysis.

Check: Record the observable result or explain the error before continuing.

Step 3 - Run the selector with the default quality floor and cost ceiling

Run the selector with the default quality floor and cost ceiling.

```
python3 select_trial.py --input trials.csv --min-auc 0.82 --max-cost 25
```

Check: Record the observable result or explain the error before continuing.

Step 4 - Calculate marginal quality gain from the third to the fifth completed trial

Calculate marginal quality gain from the third to the fifth completed trial.

Check: Record the observable result or explain the error before continuing.

Step 5 - Identify whether the best validation trial is likely over-searched and state why an untouched test set is required

Identify whether the best validation trial is likely over-searched and state why an untouched test set is required.

Check: Record the observable result or explain the error before continuing.

Step 6 - Change the cost ceiling and explain the selected-trial change using evidence

Change the cost ceiling and explain the selected-trial change using evidence.

Check: Record the observable result or explain the error before continuing.

Step 7 - Draft a final retraining configuration using the selected parameters and no tuning-only callbacks

Draft a final retraining configuration using the selected parameters and no tuning-only callbacks.

Check: Record the observable result or explain the error before continuing.

Acceptance checklist

[ ] Every required artifact exists and opens without error.

[ ] Results are tied to the supplied mock data or an approved training project.

[ ] The conclusion cites metrics, IDs or command output rather than opinion alone.

[ ] No credentials, tokens, personal data or unredacted sensitive prompts are present.

[ ] Cleanup is completed or the retained resource is documented with an owner and expiry.

Troubleshooting

Symptom	Likely cause	Corrective action
403 PERMISSION_DENIED	API disabled, wrong project, or IAM role missing	Verify gcloud config get-value project, API state and the least-privilege role.
429 RESOURCE_EXHAUSTED	Quota or rate limit reached	Stop retries, inspect quota, add exponential backoff, and ask the trainer before increasing quota.
Region/location mismatch	Dataset, bucket, model or endpoint uses another location	Recreate only the training resource in one approved region; do not copy sensitive data casually.
Result differs from expected	Data, dependency or configuration version drifted	Compare file hashes, pinned versions and parameters; retain the diff as evidence.

Cleanup

Stop local processes and remove only resources created for this activity. For live cloud work, use the Console resource views and activity evidence to verify deletion. Do not delete a shared project, dataset, bucket, registry or endpoint unless the trainer identifies it as disposable.